

The Best Open Source Automated Penetration Testing Tools

Nowadays, there is considerable amount of confusion among industry professionals and researchers regarding the significant differences between vulnerability scanning and penetration testing, as the two phrases are commonly interchanged. However, their exact technical meanings and implications are very different. A vulnerability assessment simply identifies and reports noted vulnerabilities, whereas penetration testing attempts to exploit the vulnerabilities to determine whether unauthorised access or malicious activity is possible. This article features the top five open source automated penetration testing tools.

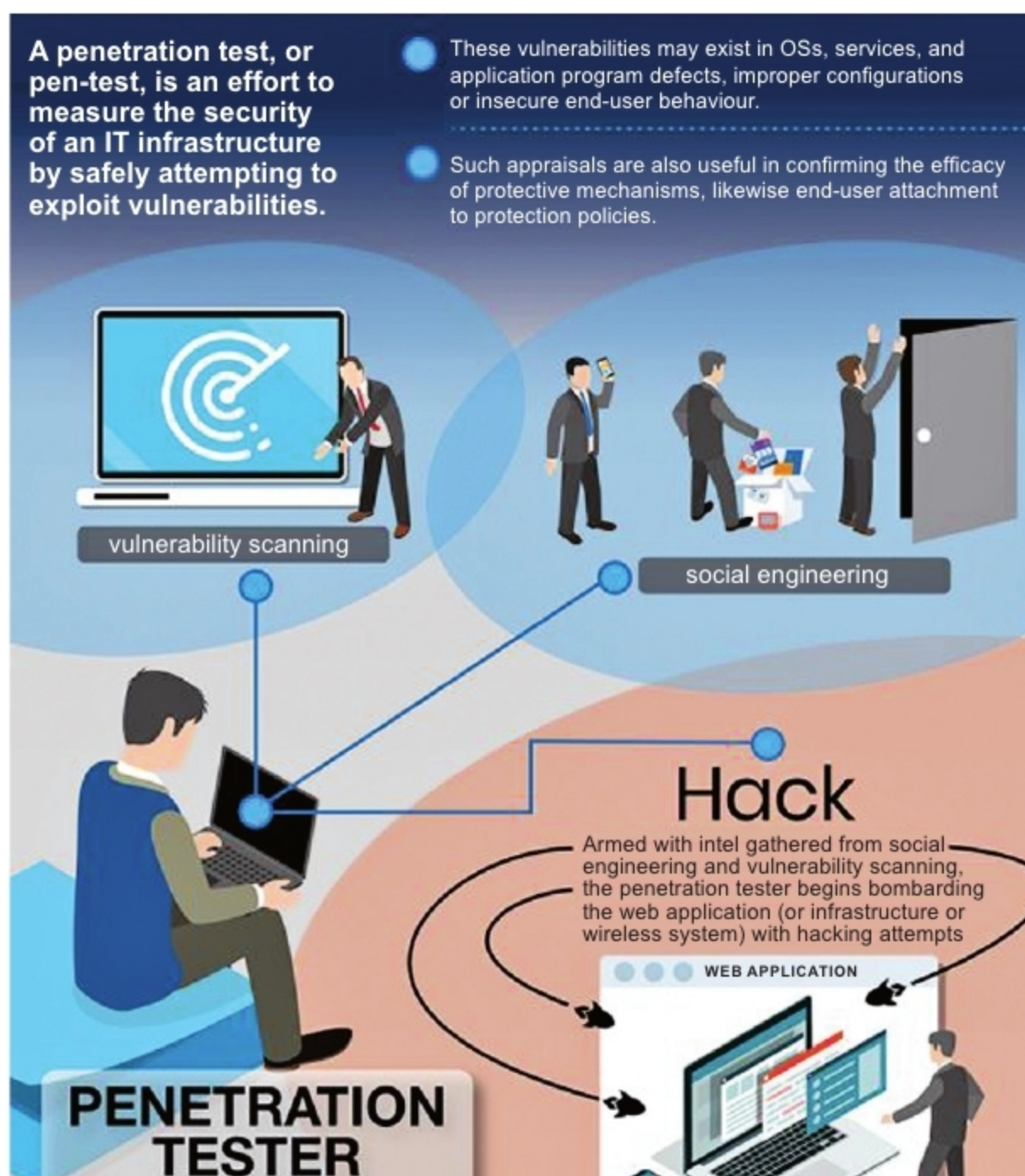


Figure 1: The work of a penetration tester

Penetration testing is a simulated cyber attack that professional ethical hackers launch to break into corporate networks to find weaknesses before attackers do. This process uses the tools and techniques available to malicious hackers. It can be automated with software applications or performed manually. In a nutshell, the process involves gathering information about the target before the test, identifying possible entry points, attempting to break in—either virtually or for real and reporting back the findings. The main objective of penetration testing is to identify security weaknesses. It can also be used to test an organisation's security policy, its adherence to compliance requirements, its employees' security awareness, and its ability to identify and respond to security incidents.

A penetration test can also highlight weaknesses in a company's security policies. For instance, although an enterprise's security policy focuses on preventing and detecting an attack on its systems, that policy may not include a process to expel a hacker. The reports generated by a penetration test provide the feedback needed by an organisation to prioritise the investments it plans